How (95)–(97) are obtained

a step-by-step companion to Sec. VI-E

every step below is checked by `analysis/vi_e_geometric_box.py`

0. What is being bounded, and why the rate

The estimator does not see the tilt angle. It fits the two-segment model to the *rate*

$$\omega_{\mathrm{nom}}(\tau) = C_1\big(\cosh C_2\tau - 1\big), \qquad C_1 = \frac{\dot M}{Wz_{CoM}}, \qquad C_2 = \sqrt{\frac{Wz_{CoM}}{J_P}}, \tag{D.1}$$

reconstructed from the gyro, and reports the onset as the argument that minimises the residual of that fit. Two questions follow, and (97) answers both:

(i) how far can the *true* rate depart from (D.1) before the fit stops describing the data? — the quantity $|e_\omega|_{\mathrm{end}}/\omega_{\mathrm{nom,end}}$;

(ii) how far does that departure move the *reported onset moment*? — the quantity $|\Delta M_{\mathrm{crit}}|$.

These are different questions with different answers, which is why the two halves of (97) carry different constants ($\frac{1}{2}$ and 1 in the first, $\frac{1}{7}$ and $\frac{1}{5}$ in the second). Section 5 answers (i) and Section 6 answers (ii). If only one line of this document is read, let it be that one.

The departure obeys the Duhamel integral (88),

$$e_\omega(\tau) = \int_0^\tau \dot h_\varphi(\tau - s)\,\rho(s)\,ds, \qquad \dot h_\varphi(u) = \frac{\cosh C_2 u}{J_P}, \tag{D.2}$$

with ρ the part of the true forcing that the estimator's span $\{1, \tau, \delta\varphi\}$ cannot absorb — the gravity remainder ρ_φ and the bilinear ground-effect coupling ρ_{GE} of (89). Everything below is about turning (D.2) into a number.

1. The crude bound, and why it is not enough

Put $|\rho(s)| \le \rho_{\max}$ inside (D.2) and pull it out:

$$|e_\omega(\tau_{\mathrm{end}})| \;\le\; \rho_{\max}\int_0^{\tau_{\mathrm{end}}} \dot h_\varphi(\tau_{\mathrm{end}} - s)\,ds \;=\; \rho_{\max}\int_0^{\tau_{\mathrm{end}}} \frac{\cosh C_2 u}{J_P}\,du \;=\; \frac{\rho_{\max}\sinh x}{J_P C_2}, \quad x \triangleq C_2\tau_{\mathrm{end}}. \tag{D.3}$$

The substitution $u = \tau_{\mathrm{end}} - s$ was used in the middle step; the integral of cosh is sinh. Using $J_P = Wz_{CoM}/C_2^2$, i.e. $J_P C_2 = Wz_{CoM}/C_2$, the inertia disappears: $|e_\omega(\tau_{\mathrm{end}})| \le \rho_{\max} C_2 \sinh x / Wz_{CoM}$. This is (90).

(D.3) is honest but blunt. It treats ρ as though it sat at its maximum for the whole window, whereas both channels start at zero at the onset and grow like τ^6 and τ^4. Evaluated over the geometric box of Sec. VI-D it returns a deviation of 446% at the slowest ramp — bigger than the signal it is supposed to bound. Sections 2–4 recover the lost factor.

2. Chebyshev's integral inequality

Statement. Let f, g be integrable on $[a, b]$. If they are *oppositely* monotone (one non-decreasing, the other non-increasing), then

$$\int_a^b fg \;\le\; \frac{1}{b-a}\int_a^b f \int_a^b g. \tag{D.4}$$

If they are monotone in the *same* direction, the inequality reverses. (Textbooks usually print the same-direction version, with $\ge$; the one needed here is the reversed one, so the direction is worth checking rather than remembering.)

Proof. Opposite monotonicity means $\big(f(u)-f(v)\big)\big(g(u)-g(v)\big) \le 0$ for every $u, v \in [a, b]$: when $u > v$ one bracket is ≥ 0 and the other ≤ 0. Integrate that product over the square $[a, b]^2$ and expand the four terms,

$$\underbrace{\iint f(u)g(u)}_{(b-a)\int fg} - \underbrace{\iint f(u)g(v)}_{\int f \int g} - \underbrace{\iint f(v)g(u)}_{\int f \int g} + \underbrace{\iint f(v)g(v)}_{(b-a)\int fg} \;=\; 2(b-a)\int fg - 2\int f\int g \;\le\; 0,$$

which is (D.4). □

Application to (D.2). Take $[a, b] = [0, \tau_{\text{end}}]$ with $f(s) = \rho(s)$ and $g(s) = \dot{h}_\varphi(\tau_{\text{end}} - s)$.

- f is non-decreasing: both channels vanish at the onset and grow monotonically over the window ($\rho_\varphi \propto \varphi^2$ with φ rising, $\rho_{GE} \propto \tau\varphi$ likewise).
- g is non-increasing *in* s: as s grows the argument $\tau_{\text{end}} - s$ shrinks and cosh decreases. Note the kernel is increasing in its own argument; it is the reflection $s \mapsto \tau_{\text{end}} - s$ that makes it decreasing in s.

They are oppositely monotone, so (D.4) applies with $\le$, which is the useful direction:

$$e_\omega(\tau_{\text{end}}) \;\le\; \underbrace{\frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} \rho}_{\bar\rho} \cdot \int_0^{\tau_{\text{end}}} \dot{h}_\varphi(\tau_{\text{end}} - s)\, ds \;=\; \frac{\bar\rho \sinh x}{J_P C_2}. \tag{D.5}$$

(D.5) is (D.3) with $\rho_{\max}$ replaced by the *time average* $\bar\rho$. Nothing else changed. All that remains is to compute $\bar\rho/\rho_{\max}$ for the two channels.

3. The reduction factors R_φ and R_{GE}: (95)

Both channels have a known shape, because φ does. From (D.1), $\varphi_{\text{nom}}(\tau) = (C_1/C_2)\Lambda(C_2\tau)$ with $\Lambda(u) = \sinh u - u$. Hence, writing $u = C_2\tau$,

$$\rho_\varphi(\tau) = \tfrac{1}{2}G''\varphi^2 = A\,\Lambda(u)^2, \qquad \rho_{GE}(\tau) = \beta_M \dot{M}\tau\,\varphi = B\,u\,\Lambda(u),$$

with A, B constants whose values never matter — they cancel in the ratio below.

Gravity channel. The maximum is at the window end, $\rho_{\varphi,\max} = A\Lambda(x)^2$. The average is, substituting $u = C_2\tau$ so that $d\tau = du/C_2$ and $\tau_{\text{end}} = x/C_2$,

$$\bar\rho_\varphi = \frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} A\Lambda(C_2\tau)^2 d\tau = \frac{C_2}{x}\cdot\frac{A}{C_2}\int_0^x \Lambda(u)^2 du = \frac{A}{x}\int_0^x \Lambda(u)^2 du,$$

so the constant A cancels and

$$R_\varphi(x) \;\triangleq\; \frac{\bar\rho_\varphi}{\rho_{\varphi,\max}} \;=\; \frac{1}{x}\,\frac{\int_0^x \Lambda(u)^2du}{\Lambda(x)^2}. \tag{D.6}$$

Ground-effect channel. Identically, $\rho_{GE,\max} = B\,x\Lambda(x)$ and

$$R_{GE}(x) \;=\; \frac{1}{x}\,\frac{\int_0^x u\,\Lambda(u)\,du}{x\,\Lambda(x)}. \tag{D.7}$$

Where 1/7 and 1/5 come from. Expand $\Lambda(u) = u^3/6 + u^5/120 + \cdots$ and keep the leading term.

Small x. $\Lambda^2 \simeq u^6/36$, so $\int_0^x \Lambda^2 \simeq x^7/(7\cdot 36)$ and

$$R_\varphi \simeq \frac{1}{x}\cdot\frac{x^7/(7\cdot 36)}{x^6/36} = \frac{1}{7}.$$

Similarly $u\Lambda \simeq u^4/6$, so $\int_0^x u\Lambda \simeq x^5/30$ and

$$R_{GE} \simeq \frac{1}{x}\cdot\frac{x^5/30}{x\cdot x^3/6} = \frac{x^4/30}{x^4/6} = \frac{1}{5}.$$

The two numbers are just the orders of the channels: a τ^6 signal averages to 1/7 of its endpoint value, a τ^4 signal to 1/5. (For τ^n the average over $[0,T]$ is $T^n/(n+1)$ against the endpoint T^n.)

Large x. $\Lambda \to \sinh u \simeq e^u/2$, so $\int_0^x \Lambda^2 \simeq e^{2x}/8$ and $R_\varphi \simeq 1/(2x)$; $\int_0^x u\Lambda \simeq xe^x/2$ and $R_{GE} \simeq 1/x$. Both fall off like $1/x$.

Why the small-x value is the supremum. Both R's decrease throughout, and this needs no numerics. Substituting $u = xt$ in the numerator rescales the window to $[0,1]$:

$$R(x) = \frac{1}{x}\frac{\int_0^x f(u)\,du}{f(x)} = \int_0^1 \frac{f(xt)}{f(x)}\,dt. \tag{D.6a}$$

In this form the small-x limit is immediate: if $f \simeq u^n$ near the origin then $f(xt)/f(x) \to t^n$ and $R \to \int_0^1 t^n dt = 1/(n{+}1)$. Now let

$$n_{\rm eff}(u) \;\triangleq\; \frac{uf'(u)}{f(u)} \;=\; \frac{d\log f}{d\log u}$$

be the *local order* of f. Differentiating the integrand of (D.6a) logarithmically,

$$\frac{\partial}{\partial x}\log\frac{f(xt)}{f(x)} = t\,\frac{f'(xt)}{f(xt)} - \frac{f'(x)}{f(x)} = \frac{n_{\rm eff}(xt) - n_{\rm eff}(x)}{x}.$$

For $t \in (0,1)$ we have $xt < x$, so if $n_{\rm eff}$ is increasing the bracket is ≤ 0: every integrand of (D.6a) decreases in x, hence so does R. □

Both channels satisfy the hypothesis, and with $\Lambda' = \cosh u - 1$ the local orders are expressible through the same $\Psi_\varphi(u) = u(\cosh u - 1)/(\sinh u - u) = u\Lambda'/\Lambda$ that Sec. 4 discarded as the wrong normaliser:

$$f = \Lambda^2: \quad n_{\rm eff} = 2\Psi_\varphi(u)\ \text{ from 6 to } \infty; \qquad f = u\Lambda: \quad n_{\rm eff} = 1 + \Psi_\varphi(u)\ \text{ from 4 to } \infty,$$

Ψ_φ being increasing with $\Psi_\varphi(0^+) = 3$. The mechanism is physical: Λ behaves as u^3 near the onset and grows exponentially later, so the larger the window the more sharply both channels are concentrated at its end, and the smaller their average is relative to the endpoint value. The small-x values are therefore suprema:

$$\bar\rho = R_\varphi(x)\,\rho_{\varphi,\max} + R_{GE}(x)\,\rho_{GE,\max}, \qquad R_\varphi \le \tfrac{1}{7}, \qquad R_{GE} \le \tfrac{1}{5}. \tag{95}$$

The suprema are approached but never attained: for $x > 0$ the inequalities are strict. Over the flown $x \in [1.79, 5.20]$ the factors run 0.089–0.133 and 0.145–0.191, so quoting $\frac{1}{7}$ and $\frac{1}{5}$ gives away 10–60% — while gaining a factor of 5 to 11 over (D.3).

4. Normalising on the rate: (96)

Divide (D.5) by the nominal rate at the same instant, $\omega_{\text{nom,end}} = C_1(\cosh x - 1)$ with $C_1 = \dot{M}/Wz_{CoM}$, and use $J_P C_2 = Wz_{CoM}/C_2$:

$$\frac{|e_\omega|_{\text{end}}}{\omega_{\text{nom,end}}} \le \frac{\bar\rho\, C_2 \sinh x / Wz_{CoM}}{(\dot{M}/Wz_{CoM})(\cosh x - 1)} = \frac{\bar\rho\, C_2}{\dot{M}} \cdot \frac{\sinh x}{\cosh x - 1}.$$

Wz_{CoM} has cancelled. Now eliminate $\dot{M}$ in favour of the moment increment the window actually spans. Since $\tau_{\text{end}} = x/C_2$,

$$\Delta M_{\text{win}} = \dot{M}\tau_{\text{end}} = \frac{\dot{M}x}{C_2} \qquad \Longleftrightarrow \qquad \frac{C_2}{\dot{M}} = \frac{x}{\Delta M_{\text{win}}},$$

and substituting,

$$\frac{|e_\omega|_{\text{end}}}{\omega_{\text{nom,end}}} \le \frac{\Psi(x)\,\bar\rho}{\Delta M_{\text{win}}}, \qquad \Psi(x) = \frac{x \sinh x}{\cosh x - 1}. \tag{96}$$

The half-angle identities $\cosh x - 1 = 2\sinh^2(x/2)$ and $\sinh x = 2\sinh(x/2)\cosh(x/2)$ collapse the quotient:

$$\Psi(x) = x\,\frac{2\sinh(x/2)\cosh(x/2)}{2\sinh^2(x/2)} = x \coth\frac{x}{2} \ \ge\ 2.$$

Ψ rises from 2 at $x \to 0$ to x as $x \to \infty$. (Had the excursion been used as the normaliser instead of the rate, the factor would have been $x(\cosh x - 1)/(\sinh x - x)$, rising from 3 — a different function, and the wrong one, since the estimator never looks at φ.)

5. Why $\Psi\bar\rho$ stays bounded: first half of (97)

Ψ grows with x and the R's shrink with it, and the two products converge to $\frac{1}{2}$ and 1. Both bounds are exact, and neither needs the series: two differentiations settle each.

The claims in integral form. Since $\Psi = x\coth(x/2)$ and $\rho_{\varphi,\max} = A\Lambda(x)^2$,

$$\Psi R_\varphi = x \coth\tfrac{x}{2} \cdot \frac{\int_0^x \Lambda^2 du}{x\Lambda(x)^2} = \coth\tfrac{x}{2}\,\frac{\int_0^x \Lambda^2 du}{\Lambda(x)^2},$$

so $\Psi R_\varphi \le \frac{1}{2}$ is exactly

$$\int_0^x \Lambda(u)^2 du \ \le\ \tfrac{1}{2}\tanh\tfrac{x}{2}\,\Lambda(x)^2, \tag{D.10}$$

and the same reduction turns $\Psi R_{GE} \le 1$ into

$$\int_0^x u\,\Lambda(u)\,du \ \le\ x\tanh\tfrac{x}{2}\,\Lambda(x). \tag{D.11}$$

The half-angle tangent is not an accident: $\tanh(x/2) = (\cosh x - 1)/\sinh x = \Lambda'(x)/\sinh x$, so both statements compare an integral of Λ against Λ and its own derivative.

Where $\frac{1}{2}$ and 1 come from. Both are the $x \to \infty$ limits, and each is one application of l'Hôpital's rule. Written as above, both products have the form $\coth(x/2)\int_0^x f/f(x)$ — with $f = \Lambda^2$ for the gravity channel and $f = u\Lambda$ for the ground-effect channel — and $\coth(x/2) \to 1$. In the remaining ratio numerator and denominator both diverge, so

$$\lim_{x\to\infty} \frac{\int_0^x f(u)\,du}{f(x)} = \lim_{x\to\infty} \frac{f(x)}{f'(x)} = \Big(\lim_{x\to\infty} \frac{f'}{f}\Big)^{-1},$$

and the two logarithmic derivatives are elementary:

$$\left.\frac{f'}{f}\right|_{f=\Lambda^2} = \frac{2\Lambda'}{\Lambda} = \frac{2(\cosh x - 1)}{\sinh x - x} \longrightarrow 2, \qquad \left.\frac{f'}{f}\right|_{f=u\Lambda} = \frac{1}{u} + \frac{\Lambda'}{\Lambda} \longrightarrow 1,$$

giving $\frac{1}{2}$ and 1. Each constant is thus the reciprocal of its kernel's exponential growth rate, which is what $\int_0^x e^{ku}du = e^{kx}/k$ says: the gravity channel gets the smaller constant because its kernel, $\Lambda^2 \sim e^{2x}/4$, is the square of the other's, $u\Lambda \sim u\, e^x/2$.

The only theorem used. Everything below rests on one corollary of the mean value theorem, applied three times and nowhere supplemented.

Lemma. If f is continuous on $[0, b]$, differentiable on $(0, b)$ with $f' \geq 0$ there, and $f(0) = 0$, then $f \geq 0$ on $[0, b]$. *Proof:* for $0 \leq s < t \leq b$ the mean value theorem gives $f(t)-f(s) = f'(\xi)(t-s) \geq 0$, so f is non-decreasing; taking $s = 0$ gives $f(t) \geq f(0) = 0$. □

Note what is *not* used. No limit as $x \to \infty$ enters: the constant $\frac{1}{2}$ comes from the algebra of (D.10), and the inequality is established at each x separately. No monotonicity of ΨR_φ is assumed or proved. The limits quoted above serve only to show the constants cannot be improved.

Proof of (D.10). Let $D(x)$ be the right side minus the left, so that (D.10) is $D \geq 0$. Every quantity divided by below — $\coth(x/2)$, Λ^2, Λ, $\cosh^2(x/2)$ — is strictly positive for $x > 0$, so each step is an equivalence.

$D(0) = 0$, since $\tanh 0 = \Lambda(0) = \int_0^0 = 0$. Differentiating and factoring out $\Lambda > 0$,

$$D'(x) = \Lambda(x)\, B(x), \qquad B(x) = \tfrac{1}{4}\operatorname{sech}^2 \tfrac{x}{2}\, \Lambda + \tanh \tfrac{x}{2}\, \Lambda' - \Lambda,$$

so $D' \geq 0$ iff $B \geq 0$. Write $y = x/2$, $S = \sinh y$, $C = \cosh y$, giving $\Lambda = 2SC - 2y$ and $\Lambda' = 2S^2$. Multiplying B by $2C^2 > 0$,

$$2C^2 B = (SC - y) + 4S^3C - 4SC^3 + 4yC^2,$$

and since $S^3C - SC^3 = SC(S^2 - C^2) = -SC$ the two cubic terms collapse to $-4SC$, leaving $y(4C^2 - 1) - 3SC = y(3 + 4S^2) - 3SC$. Now $4S^2 = 2\cosh x - 2$ and $2SC = \sinh x$ and $y = x/2$, so

$$B(x) = \frac{\phi(x)}{4\cosh^2(x/2)}, \qquad \phi(x) \triangleq x + 2x\cosh x - 3\sinh x,$$

and $B \geq 0$ iff $\phi \geq 0$. Finally

$$\phi'(x) = 1 - \cosh x + 2x\sinh x, \qquad \phi''(x) = \sinh x + 2x\cosh x.$$

First application. $f = \phi'$: both terms of $\phi'' = \sinh x + 2x\cosh x$ are non-negative for $x \geq 0$, and $\phi'(0) = 1 - 1 + 0 = 0$. The Lemma gives $\phi' \geq 0$.

Second application. $f = \phi$: $\phi' \geq 0$ by the first application and $\phi(0) = 0$. The Lemma gives $\phi \geq 0$, hence $B \geq 0$, hence $D' = \Lambda B \geq 0$.

Third application. $f = D$: $D' \geq 0$ by the second application and $D(0) = 0$. The Lemma gives $D \geq 0$, which is (D.10). □

Eq. (D.11) follows the same way, its difference having $E(0) = 0$ and $E' \geq 0$; alternatively its power series has the closed general coefficient $\left[2^{m-1}/m + (m^2 - 6m + 2)/3\right]/(m-1)!$ with $m = 2n$, which vanishes at $m = 2, 4$ and is positive for $m \geq 6$ because both terms are.

No monotonicity of the products is claimed or needed — only that they do not exceed their limits, which is what (D.10) and (D.11) say. Expanding $\bar{\rho}$ by (95),

$$\Psi\bar{\rho} = \big(\Psi R_\varphi\big)\rho_{\varphi,\max} + \big(\Psi R_{GE}\big)\rho_{GE,\max} \;\leq\; \tfrac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max},$$

which is the left half of (97). *x has disappeared* — and with it C_2, J_P and Wz_{CoM}. This is the whole point of the exercise: the bound no longer depends on the window length, so it does not have to be solved for, only bounded well enough to keep $\rho_{GE,\max}$ finite (which is what (92)–(93) do).

6. From the rate deviation to the threshold: second half of (97)

The reported quantity is not e_ω but the onset moment, so the deviation must be pushed one step further.

How the onset responds to a perturbation. The estimator picks t_0 to minimise $J(t_0) = \int \big(y(t) - \hat\omega(t;t_0)\big)^2 dt$ with $\hat\omega(t;t_0) = C_1(\cosh C_2(t-t_0) - 1)$. Write the data as the nominal solution plus the deviation, $y = \hat\omega(t;t_0^*) + e_\omega$, and the minimiser as $t_0 = t_0^* + \delta$. To first order $\hat\omega(t;t_0^*+\delta) \simeq \hat\omega(t;t_0^*) + \delta\,\partial_{t_0}\hat\omega$, so the residual is $e_\omega - \delta\,\partial_{t_0}\hat\omega$ and stationarity $\partial J/\partial t_0 = 0$ reads

$$\int \big(e_\omega - \delta\,\partial_{t_0}\hat\omega\big)\partial_{t_0}\hat\omega\,dt = 0 \qquad \Longrightarrow \qquad \delta = \frac{\langle e_\omega,\ \partial_{t_0}\hat\omega\rangle}{\|\partial_{t_0}\hat\omega\|^2}.$$

With $\partial_{t_0}\hat\omega = -C_1C_2\sinh(C_2\tau)$ and $\Delta M_{\rm crit} = \dot M\delta$, and using $C_1 = \dot M/Wz_{CoM}$,

$$\Delta M_{\rm crit} = -\,\frac{Wz_{CoM}\,\langle e_\omega,\ \sinh(C_2\tau)\rangle}{C_2\,\|\sinh(C_2\tau)\|^2}. \tag{D.8}$$

This is exactly the reduction `error_budget.py` implements.

It is an average of ρ with a non-negative weight. Substitute the Duhamel integral (D.2) into (D.8) and exchange the order of integration (Fubini; both integrands are continuous on a bounded domain):

$$\int_0^T \sinh(C_2\tau)\int_0^\tau \dot h_\varphi(\tau-s)\rho(s)\,ds\,d\tau = \int_0^T \rho(s)\underbrace{\left[\int_s^T \sinh(C_2\tau)\,\dot h_\varphi(\tau-s)\,d\tau\right]}_{\text{depends on } s \text{ only}} ds,$$

the inner limits changing from $s\in[0,\tau]$ to $\tau\in[s,T]$. Hence

$$\Delta M_{\rm crit} = -\int_0^T \rho(s)\,w(s)\,ds, \qquad w(s) = \frac{Wz_{CoM}\displaystyle\int_s^T \sinh(C_2\tau)\cosh\big(C_2(\tau-s)\big)d\tau}{J_PC_2\|\sinh(C_2\tau)\|^2}. \tag{D.9}$$

Three properties of w finish the argument, and all three are checked in code (`error_budget.py --check`):

1. $w \ge 0$, because the integrand is a product of two hyperbolic functions of non-negative arguments;
2. $\int_0^T w = 1$, exactly. This is not a numerical coincidence and is worth doing, because it is what lets $\bar\rho$ enter (97) bare. Integrate the numerator of (D.9) over s and exchange the order: the region $0 \le s \le \tau \le T$ is the same triangle read either way, so

   $$\int_0^T\!\!\int_s^T \sinh(C_2\tau)\cosh\big(C_2(\tau-s)\big)d\tau\,ds = \int_0^T \sinh(C_2\tau)\left[\int_0^\tau \cosh\big(C_2(\tau-s)\big)ds\right]d\tau.$$

   The inner integral is elementary — substitute $u = \tau - s$, whose sign flip is cancelled by the reversed limits —

   $$\int_0^\tau \cosh\big(C_2(\tau-s)\big)ds = \int_0^\tau \cosh(C_2u)\,du = \frac{\sinh(C_2\tau)}{C_2},$$

   and putting it back reproduces the very norm that sits in the denominator of (D.9),

   $$\frac{1}{C_2}\int_0^T \sinh^2(C_2\tau)\,d\tau = \frac{\|\sinh(C_2\tau)\|^2}{C_2}.$$

Hence

$$\int_0^T w = \frac{W z_{CoM} \, \| \sinh \|^2 / C_2}{J_P C_2 \| \sinh \|^2} = \frac{W z_{CoM}}{J_P C_2^2} = 1$$

by $J_P C_2^2 = W z_{CoM}$ — the same identity that removes the inertia everywhere else in this document. Neither the window length nor any constant survives. `error_budget.py --check` evaluates (D.9) at $\rho \equiv 1$ and returns -1.0005 to -1.0001 across five operating points, which is the quadrature error on an identity rather than evidence for it. The reading is that a uniform unmodelled moment displaces the identified threshold one-for-one, and now that is a consequence rather than an assumption;

3. w is non-increasing in s — as s grows both the integration range and the cosh factor shrink (verified numerically over the window).

Property 1 plus 2 alone would give $|\Delta M_{\text{crit}}| \le \rho_{\max}$. But ρ is increasing and w is decreasing, so *Chebyshev applies a second time*, now to (D.9):

$$\Big| \int_0^T \rho \, w \Big| \;\le\; \frac{1}{T} \int_0^T \rho \int_0^T w \;=\; \bar{\rho} \cdot 1 \;=\; \bar{\rho},$$

and substituting (95),

$$|\Delta M_{\text{crit}}| \;\le\; \bar{\rho} \;\le\; \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}, \tag{97b}$$

the right half of (97).

Why the constants differ from the left half. The left half asks how far the rate curve is displaced, so the kernel integral Ψ multiplies $\bar{\rho}$ and the products ΨR climb to $\frac{1}{2}$ and 1. The right half asks how far the *minimiser* moves, and the minimiser is found by correlating against $\sinh(C_2 \tau)$ — a normalised operation, $\int w = 1$, with no kernel gain left over. So there $\bar{\rho}$ enters bare, and the constants are the R's themselves, $\frac{1}{7}$ and $\frac{1}{5}$. The threshold bound is the stronger of the two by roughly a factor of four, which is why it, and not the relative rate bound, is the number carried into the offset budget.

7. A worked example, checkable by hand

Take the roll axis at the fastest ramp, evaluated over the geometric box of Sec. VI-D rather than at any fitted constant: $W = 31.59$ N (fully ballasted, $m = 3.221$ kg), $z_{CoM} = 0.30$ m, $a = l_p + p_{\text{off}} = 0.160$ m, $\dot{M} = 1.20$ N m/s, $\varphi_{\max} = 10° = 0.17453$ rad, $\beta_M = 0.0345$ rad^{-1}, $J_{CoM}^{\text{CAD}} = 0.0537$ kg m^2.

step	expression	value
geometry	$Wz_{CoM} = W z_{CoM}$	9.477 N m
(82) lower	$J_P \geq m(z_{CoM}^2 + a^2)$	0.3724 kg m^2
(82) upper	$J_P \leq J_{CoM}^{\mathrm{CAD}} + m(z_{CoM}^2 + a^2)$	0.4261 kg m^2
	$C_2 \leq \sqrt{g z_{CoM}/(z_{CoM}^2 + a^2)}$	5.045 s^{-1}
tilt cap $\to \Lambda$	$\Lambda_{\max} = \varphi_{\max} W z_{CoM} C_2/\dot{M}$	6.954
(92)	$\bar{x} = (6\Lambda_{\max})^{1/3}$	3.468
exact	solve $\sinh x - x = \Lambda_{\max}$	$x = 2.993$
(93)	$\tau_{\mathrm{end}} \leq (6\varphi_{\max} J_P/\dot{M})^{1/3}$	0.719 s
(93)	$\Delta M_{\mathrm{win}} = \dot{M}\tau_{\mathrm{end}}$	0.863 N m
channel	$\rho_{\varphi,\max} = \frac{1}{2} W a\, \varphi_{\max}^2$	76.98 mN m
(93)	$\rho_{GE,\max} = \lvert\beta_M\rvert \Delta M_{\mathrm{win}} \varphi_{\max}$	5.20 mN m
(D.6)	$R_\varphi(3.468)$	0.1125
(D.7)	$R_{GE}(3.468)$	0.1705
(95)	$\bar{\rho} = R_\varphi \rho_{\varphi,\max} + R_{GE}\rho_{GE,\max}$	9.54 mN m
	(against $\rho_{\max} = 82.2$ mN m)	
(96)	$\Psi(\bar{x}) = \bar{x}\coth(\bar{x}/2)$	3.692
(96)	$\Psi\bar{\rho}/\Delta M_{\mathrm{win}}$	4.08%
(97)	$(\frac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max})/\Delta M_{\mathrm{win}}$	5.06%
(97b)	$\rho_{\varphi,\max}/7 + \rho_{GE,\max}/5$	12.04 mN m
	$\div W_{\min}$	0.400 mm
(D.3)	envelope only, $\Psi\rho_{\max}/\Delta M_{\mathrm{win}}$	35.2%

Two readings. First, (97) costs about a quarter over (96) — the price of dropping x from the statement — whereas the envelope form costs a factor of nine. Second, this is the *worst admissible* case: the box is evaluated at the 10° tilt cap, at the top of the z_{CoM} range, and with the inertia at whichever end of (82) is unfavourable for the quantity being bounded. Over the runs actually flown the excursions are 5–9° and the exact per-run propagation, which convolves the measured ρ instead of any envelope, gives 0.04 mN m in the median and 0.15 at worst — 0.001 to 0.004 mm of offset.

8. Summary of the chain

	step	what it buys
(D.2)	$e_\omega = \int \dot{h}_\varphi(\tau_{\mathrm{end}} - s)\rho(s)ds$	the exact deviation
(D.3)	$\lvert\rho\rvert \leq \rho_{\max}$, integrate the kernel	(90); too weak by $\sim 8\times$
(D.4)	Chebyshev, $\rho\uparrow$ against $\dot{h}_\varphi(\tau_{\mathrm{end}} - \cdot)\downarrow$	$\rho_{\max} \to \bar{\rho}$
(D.6–7)	known shapes Λ^2 and $u\Lambda$	the ratios $\bar{\rho}/\rho_{\max}$
(D.6a)	$R = \int_0^1 f(xt)/f(x)dt$, $n_{\mathrm{eff}}\uparrow$	(95): $R_\varphi \leq \frac{1}{7}$, $R_{GE} \leq \frac{1}{5}$
(96)	divide by $\omega_{\mathrm{nom,end}}$, use $\Delta M_{\mathrm{win}} = \dot{M}x/C_2$	$\Psi = x\coth(x/2)$; Wz_{CoM}, J_P gone
	$\Psi R_\varphi \uparrow \frac{1}{2}$, $\Psi R_{GE} \uparrow 1$	(97) left half; x gone
(D.8–9)	linearise the onset, Fubini, $w \geq 0$, $\int w = 1$	$\Delta M_{\mathrm{crit}} = -\langle\rho\rangle_w$
	Chebyshev again, $\rho\uparrow$ against $w\downarrow$	(97) right half: $\frac{1}{7}, \frac{1}{5}$

Chebyshev is used twice, for the same reason both times: the forcing is concentrated at the end of the window and every kernel in the problem weights the end of the window least. The two uses are independent — the first converts a kernel integral, the second a normalised correlation — which is why the constants they leave behind differ.